

Quantifying Generative AI's Impact on Workforce Skill Demands: Evidence from Agriculture and Sustainability Job Postings

Sadhana Vasanthakumar · Vinayak Ghai · Ziyong Qu · Nikhil Srinivas

Faculty Advisor: Dr. Yanhong Jin, Department of Agricultural, Food, and Resource Economics, Rutgers University

Introduction

The job market is changing faster than the curriculum.

Generative AI is reshaping what employers ask for before anyone is hired. Applicant Tracking Systems (ATS) now screens resumes automatically — meaning the wrong keywords eliminate qualified candidates before any human review.

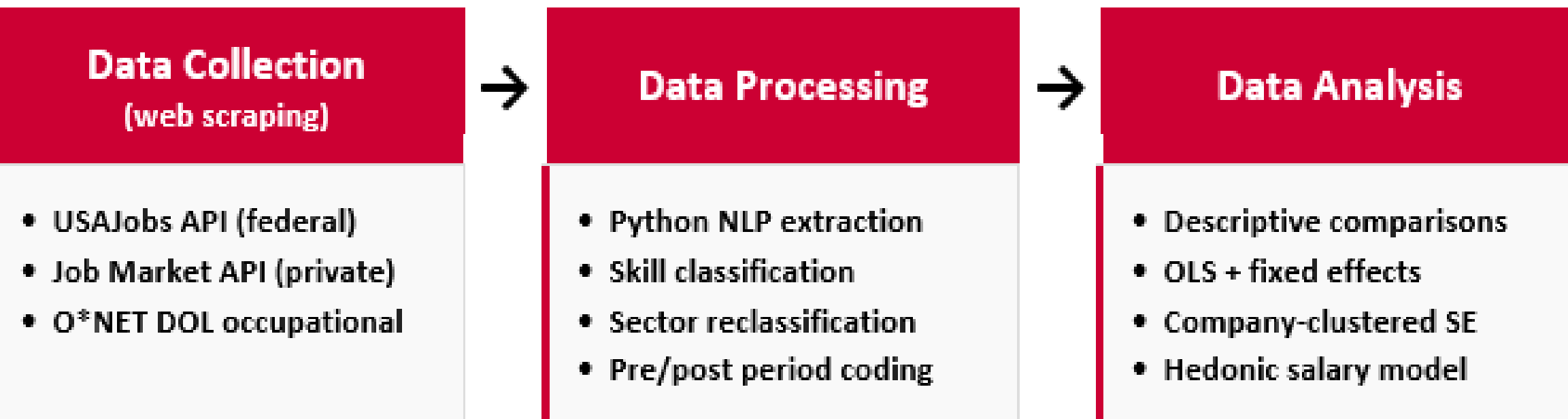
Agriculture and sustainability sectors are especially exposed: both face rapid technological transformation and a workforce research base that largely ignores entry-level, sector-specific dynamics.

We analyzed **25,516 entry-level job postings** across federal, private sector, and occupational data sources — split by ChatGPT's November 2022 release — to answer a question that matters directly to students and the institutions preparing them: ***what does the labor market demand right now, and whom is it leaving out?***

Our goal is to identify:

- How technical, digital, and soft skill requirements shifted after ChatGPT
- Whether a compensation gap exists across sectors and whether it has been affected by GenAI.
- What structural barriers do undergraduates face when entering these fields, and how can universities respond

Data & Methods



Temporal Framework



A Systems of Equations

$$Skill_j = \beta_0 + \beta_1(After_ChatGPT) + \beta_2(Sector) + \beta_3(Location) + \beta_4(Job_Type) + \epsilon_j$$

where $j \in \{Technical, Digital, Soft\}$

$$\log(Salary) = \beta_0 + \beta_1(After_ChatGPT) + \beta_2(Sector) + \beta_3(Sector \times After) + \beta_4(Grade) + \alpha_i + \gamma^t + \delta^R + \epsilon$$

 α_i = company FE γ^t = year FE δ^R = location FE

Dataset Overview

Total Jobs Analyzed	25,516
Agriculture Sector	2,661 (10.4%)
Sustainability Sector	22,855 (89.6%)
Agencies/ Employers	105
States Represented	50 + D.C.

Results

Federal hiring in agriculture and sustainability declined meaningfully in the post-GenAI period — consistent with documented government-wide hiring contractions. Yet within that shrinking pool, what employers demand is shifting: 86 new technologies entered occupation requirements between 2020 and 2024, with growing signals that technical certifications and domain credentials are becoming baseline screening criteria.

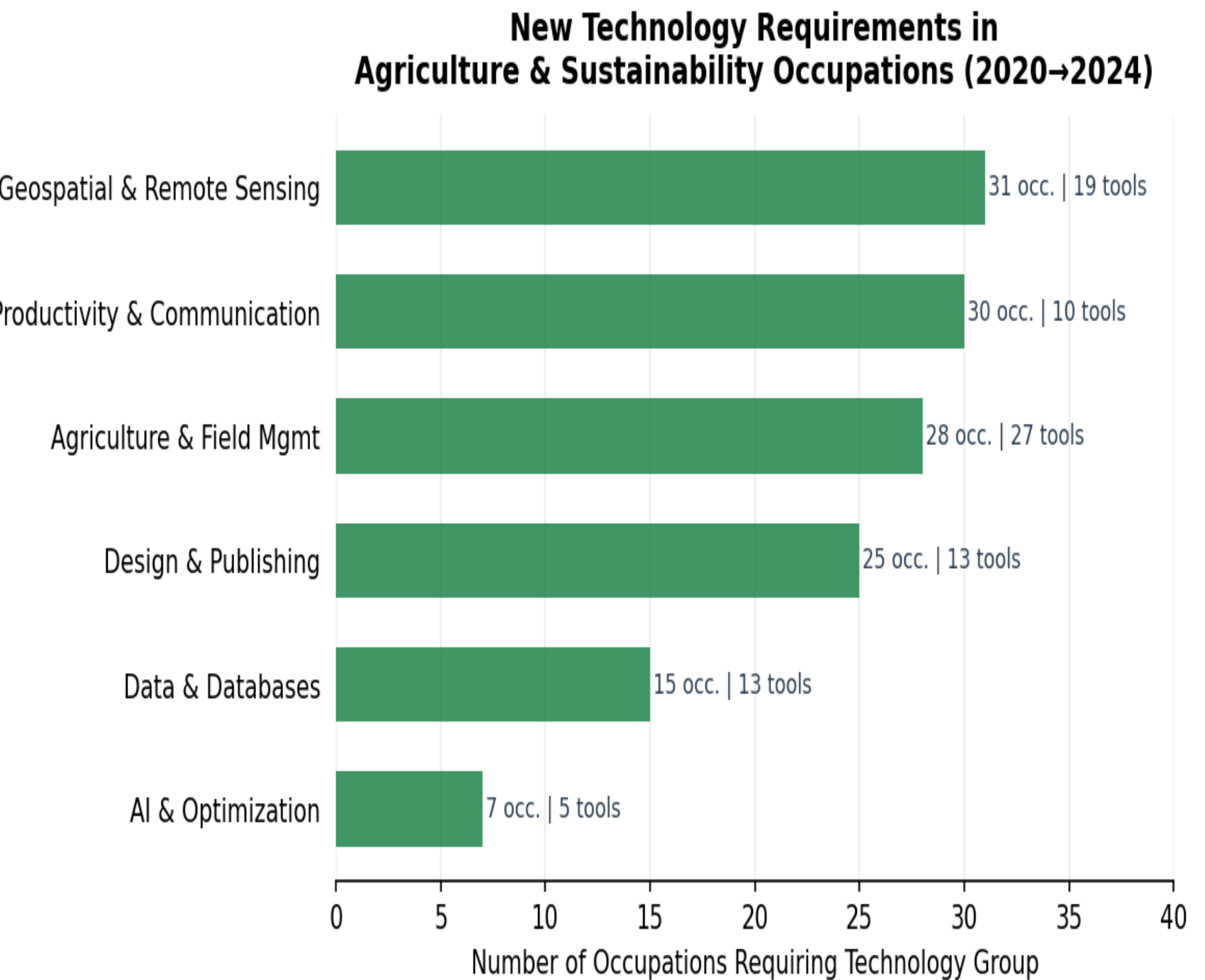
Across all periods and both sectors, soft skills remain dominant — and as technical demands expand, adaptability and active learning emerge as the only skill categories with a net positive trajectory.

Skill & Workforce Evolution

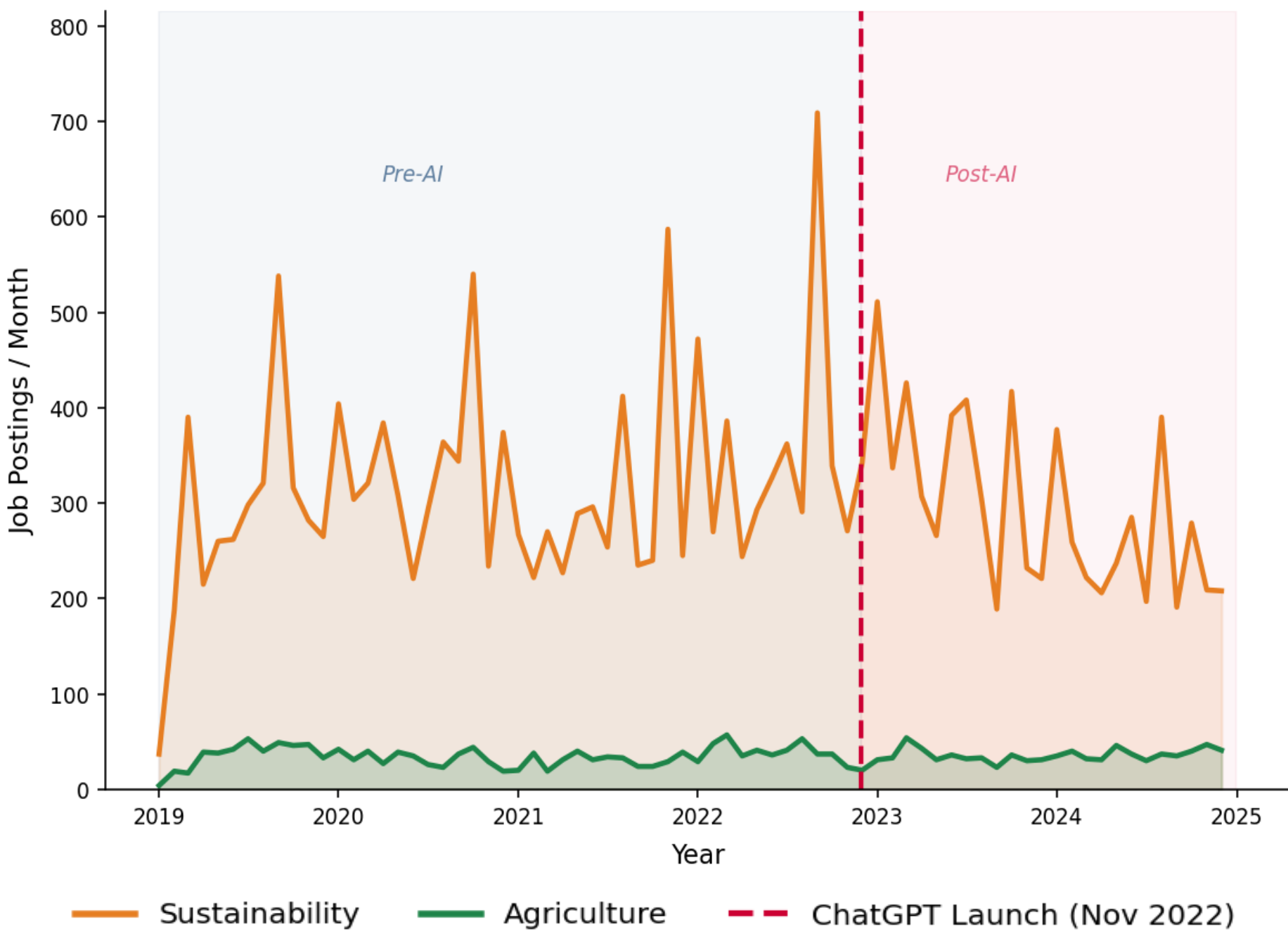
- Soft skills dominate both periods
- Learning and adaptability is the only skill category with a net positive shift
- Data science and GIS roles are growing; wildland fire management is declining
- 86 new technologies entered occupation requirements, signaling a rising technical floor

Compensation and Sector Findings

- The apparent salary gap between sectors disappears once GS grade and location are controlled
- Federal GS (General Schedule) grades determine pay bands — GS-7 and GS-9 carry a 6.8%, 13.2% premium over GS-5, respectively.
- Sustainability roles require soft skills at 5× the rate of agriculture in the private sector



Agriculture & Sustainability Hiring Trend Before and After ChatGPT (2019-2024)



Soft	Digital	Technical
Adaptability	Data Analytics	Compliance
Management and Leadership	Geospatial Tech (GIS/Remote Sensing)	Renewable Energy
Communication	Python/R Programming	ESG Reporting
Organizational Planning	Precision Ag Tech (Automation / IoT)	Equipment Maintenance
Complex Problem Solving	Statistical Modeling	Carbon Accounting

Skill Importance Shifts in Agriculture & Sustainability Occupations (O*NET 2020–2024)



Entry Level Barriers for Undergraduates

- GIS credentials now required across 13 of 21 occupation categories
- Applicants without GIS or data analytics training face growing screening barriers for roles that did not require these credentials when current seniors enrolled
- Federal early-career rate is 3.4% vs 32% in the private sector, a structural difference in how federal agencies onboard entry-level workers

Conclusions

As technical software demands rise across both federal and private sector roles, employers increasingly signal a preference for candidates who actively learn and adapt. Prior experience or certification in emerging tools — particularly GIS, remote sensing, and data analytics — meaningfully improves screening outcomes for early-career applicants in agriculture and sustainability.

Active Learning & Adaptability

86

NEW TECHNOLOGIES
requirements since 2020

13 of 21

Occupations now
require GIS credentials +
certifications

3.4% vs 32%

EARLY-CAREER POSTINGS
FEDERAL vs. PRIVATE

Policy Implications

For Students: Build skills in GIS, data analytics, and adaptability. Optimize resumes for ATS screening and pursue federal Pathways programs or private sector internships early.

For Universities: Embed GIS and data literacy across agricultural programs, integrate ATS navigation into career services, and create structured pathways to internship access before students enter the job market.

Future Direction

Expanding NLP extraction across larger and more diverse datasets, including additional private sector sources, would substantially strengthen these findings. Future work should examine task-level changes using transformer-based models to detect shifts in what workers are asked to do beyond credential listing. Longitudinal tracking of GIS and data requirements as current cohorts graduate would test whether the screening barriers identified here translate into measurable employment outcomes.

References

- Eloundou et al. (2023), "GPTs are GPTs," arXiv
- Nayga Jr. et al. (2024), "The looming demographic cliff," AJAE
- Wilson et al. (2019), "What are employers looking for?" NACTA Journal

Acknowledgements

We gratefully acknowledge Dr. Yanhong Jin for her guidance and mentorship, and the Rutgers Aresty Research Center for supporting this work.



RUTGERS